

The project I decided to create is an interactive solar system using WebGL exclusively. All eight planets, and the sun are rendered as 3D spheres orbiting in real-time. They each have their own orbital speed which helps to better convey the simulation of the system. The camera can be moved around the scene using the mouse, as well as the ability to zoom in and out using the scroll wheel.

GLSL and fragment shaders, were the main techniques used in this project. Each planet is rendered using the same mesh, but with a different MVP matrix which positions and scales them in the correct position of the scene. The mat4 math library was used for the projection matrices, as well as the lookAt function for the camera and perspective function.

The biggest challenge I had was the matrix math for WebGL, since I am decent at matrix multiplication, but when multiple matrices involved it can get confusing. This was vital for the functions involved in my project to work. Overall this project was a fun challenge, I wish I could have had more time to add more lighting and even some stars to the scene, but my time was rather limited.